

QCT Supplementary Material 0

Proof Map and Index

Complete Index to All Named Results Across Eight Standalone Modules

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1 Module Overview

This document is the proof map and index for the QCT Supplementary series of eight standalone modules. Each module is a self-contained published document; this map provides a unified index to all 171 named results (theorems, lemmas, propositions, corollaries, definitions, remarks, conjectures) across the series.

Module Files

- M1 QCT-Supp-1-Postulates-v1** Postulates P1/P2/P3, Classical Mechanics, General Relativity, Quantum Mechanics. 37 pages.
- M2 QCT-Supp-2-Identification-v1** Structural Identifications ID1, ID2, ID3; Fine Structure Constant α ; SC1 theorem. 37 pages.
- M3 QCT-Supp-3-MassGap-v1** Yang–Mills Existence and Mass Gap (Clay Prize); Appendices YM and T5; Theorems U1 and NS. 78 pages.
- M4 QCT-Supp-4-ElectronMass-v1** Hopf Solitons; $N_c = 20$; Electron Mass Formula; Appendices G1 and MF. 26 pages.
- M5 QCT-Supp-5-Cosmology-v1** Cosmological derivations C1–C11; Dark Matter and Dark Energy. 26 pages.
- M6 QCT-Supp-6-Spectral-v1** Electromagnetism; Force Mechanism; Weinberg Angle; Key Predictions. 45 pages.

M2+ QCT-Supp-2-Identification-v1 Also contains Appendix CLOSE: Final Closure Proofs (19 April 2026); moved from former M7.

M8 QCT-Supp-7-Chemistry-v1 Chemical and Molecular Predictions: CH1–CH5, Predictions P53–P62. Active.

M9 QCT-Supp-8-ParticleMasses-v1 Particle Mass Spectrum: Proton ($m_p/m_e = 6\pi^5$), Muon ($m_\mu/m_e = 20\pi^3/3$). Active.

M10 QCT-Supp-9-NuclearPhysics-v1 Nuclear Physics: binding energies, hadron structure, strong confinement. Structured shell; content in preparation.

M11 QCT-Supp-10-Neutrinos-v1 Neutrino Physics: masses, mixing angles, oscillation parameters. Structured shell; content in preparation.

M12 QCT-Supp-11-QuantumGravity-v1 Quantum Gravity and Planck Scale: black holes, Hawking radiation, graviton. Structured shell; content in preparation.

M13 QCT-Supp-12-DarkMatter-v1 Dark Matter: $Q > 1$ soliton candidates, mass predictions, cosmological fraction. Structured shell; content in preparation.

Status Legend

Proved Complete proof within QCT framework; no residual gaps.

— Status to be read from module (may be proved, near-theorem, or insight).

Def Foundational definition.

Rem Remark or observation.

Open Stated as conjecture; proof incomplete.

2 Dependency Chain

1. **M1** establishes P1, P2, P3 and derives fundamental scales (c , $\hbar$, G), Classical Mechanics, General Relativity, Quantum Mechanics. No prior modules needed.
2. **M2** uses P1/P2/P3 to prove SC1 ($d = 5$), ID1 (moduli space = D_5), ID2 (soliton = Rac), ID3 (signature $(5, 2)$), and G -equivariance. Together these establish $\alpha^{-1} = 137.036$ from the postulates.
3. **M4** uses M2 (ID1, ID2, SC1) to prove $N_c = 20$ and the electron mass formula $m_e = 20m_P(\pi/4) \exp(-11\pi^2/2)$.
4. **M3** uses M2 (structural identifications) and thermodynamic analysis to prove Yang–Mills existence and the mass gap $\Delta_{YM} = m_e > 0$.
5. **M5** uses M4 ($N_c = 20$, Hopf instanton action) to derive $\Omega_\Lambda = 0.704$, $\eta = 6.18 \times 10^{-10}$, $g_s^* = 11$, $\Omega_{\text{CDM}}/\Omega_b = 5.41$.
6. **M6** uses M2 (Hopf connection, D_5 geometry) for EM; derives the Weinberg angle via spectral zeta on D_5 ; tabulates 12 numerical predictions.

7. **M2** (updated) also contains Appendix CLOSE: final closure proofs for $N_c = 20$, $C11$, $\hbar = 2\varepsilon\tau$, spin-statistics, Weinberg angle corrections (19 April 2026).
8. **M8** uses M2, M4 to derive chemistry theorems CH1–CH5 and predictions P53–P62.
9. **M9** uses M2, M4, M3 to derive the proton mass formula $m_p/m_e = 6\pi^5$ and muon mass formula $m_\mu/m_e = 20\pi^3/3$.
10. **M10–M13** are structured shells with content in preparation: Nuclear Physics, Neutrino Physics, Quantum Gravity, Dark Matter. Derivations from P1/P2/P3 are planned for future publications.

3 Index of Named Results

Name / Title	Type	Label	Mod	Status
P1 — Full Formal Statement	Definition	—	M1	Def
P2 — Full Formal Statement	Definition	—	M1	Def
P3 — Full Formal Statement	Definition	—	M1	Def
Fundamental Scales	Theorem	thm:scales	M1	Proved
Adjacency is not a partial order	Proposition	prop:adj-not-po	M1	Proved
Tiling Completeness	Theorem	thm:tiling-completeness	M1	Proved
Open stars are open	Lemma	lem:stars-open	M1	Proved
Partition by open cell interiors	Lemma	lem:partition	M1	Proved
(definition)	Definition	—	M1	Def
(definition)	Definition	—	M1	Def
Points Cells	Theorem	thm:points-cells	M1	Proved
(remark)	Remark	—	M1	Rem
Equivalence Principle	Theorem	—	M1	Proved
Gravitational Time Dilation	Theorem	—	M1	Proved
Light Deflection	Theorem	—	M1	Proved
Gravitational Redshift	Theorem	—	M1	Proved
Event Horizon	Theorem	—	M1	Proved
Gravitational Wave Modes	Theorem	—	M1	Proved
Perihelion Precession	Theorem	—	M1	Proved
Minimum Soliton Mass	Theorem	—	M1	Proved
Friedmann Equation	Theorem	—	M1	Proved
(theorem)	Theorem	—	M6	—
(theorem)	Theorem	—	M6	—
Spectral-Geometric Consistency (SC1)	Self-Theorem	thm:sc1	M2	Proved
Status of the two requirements	Remark	—	M2	Rem
SC1 as self-consistency	Remark	—	M2	Rem
from SC1	Corollary	cor:sc1-nc	M2	Proved
Robustness: gives the same solution	Remark	—	M2	Rem
Direct branching obstruction	Lemma	lem:branching	M2	—
3+2 split from P1 and SC1	Lemma	lem:3plus2	M2	Proved

Name / Title	Type	Label	Mod	Status
Truncation at from the Joseph ideal	Lemma	lem:joseph-truncation	M2	Proved
Hopf doubling from	Lemma	lem:hopf-doubling	M2	Proved
from P1/P2/P3 ($N_c = 20$)	Theorem	thm:nc20-proved	M2	Proved
Top-Sector Equivariance	Theorem	thm:top-sector	M2	Proved
	Corollary	cor:nc20	M2	—
from top-sector embedding	Corollary	cor:nc20-formal	M2	—
rem:nc20-remark	Remark	rem:nc20-remark	M2	Rem
-equivariance of the coupling vertex	Theorem	thm:g-equiv	M2	Proved
	Corollary	cor:alpha-wyler	M2	Proved
Consistency checks	Remark	—	M2	Rem
Status of	Remark	—	M2	Rem
-type identity, all orders	Theorem	thm:ktype-identity	M2	Proved
Discrete decomposition	Proposition	prop:discrete-decomp	M2	—
Uniqueness by induction	Proposition	prop:uniqueness	M2	—
Completeness	Proposition	prop:completeness	M2	—
Flato–Fronsdal for	Theorem	thm:ff-decomp	M2	Proved
cor:photon-mult1	Corollary	cor:photon-mult1	M2	Proved
Multiplicity-free	Corollary	cor:multiplicity-free	M2	Proved
Intrinsic split via Howe duality	Conjecture	conj:howe	M2	Open
Three orientation moduli from P1	Lemma	lem:orient-3	M2	—
Axial symmetry of the classical minimise	Remark	rem:axial-symmetry	M2	Rem
(remark)	Remark	—	M2	Rem
One Hopf phase modulus	Lemma	lem:phase-1	M2	—
(remark)	Remark	—	M2	Rem
One scale modulus from P1/P2	Lemma	lem:scale-1	M2	—
(remark)	Remark	—	M2	Rem
SO(3) branching rule and singlet identif	Lemma	lem:so3-branch	M2	—
generates Thomas precession on the Hopf	Lemma	lem:thomas-m46	M2	—
generates scale change; is the scale dir	Lemma	lem:scale-m56	M2	—
Confidence assessment for ID3	Remark	—	M2	Rem
Parity is a QCT Vacuum Symmetry	Lemma	lem:parity-vacuum	M2	—
Parity Induces -Equivariant Anti-holomorp	Lemma	lem:parity-antiholo-lemma	M2	—
G-transitivity on	Proposition	prop:transitivity	M2	—
BPS moduli space is Kähler	Proposition	prop:bps-kahler	M2	—
Hopf Fibration is CR-Holomorphic; Moduli	Lemma	lem:hopf-kahler	M2	—

Name / Title	Type	Label	Mod	Status
Gap statement and corrected logical step	Remark	rem:hopf-kahler-gaps	M2	Rem
Lorentz Complexification of Hopf Moduli	Proposition	prop:lorentz-complexification	M2	—
SL(2,C) formalisation of QCT orientation	Lemma	lem:sl2c-formal	M2	—
Rapidity vs velocity in the Harish-Chand	Remark	rem:rapidity-velocity	M2	Rem
Unique determination of	Corollary	cor:D5-unique	M2	—
Gap statement	Remark	rem:lorentz-gap	M2	Rem
-type identification of the ground state	Lemma	lem:ktype-rac	M2	—
vs : topological vs representation-theor	Remark	rem:Q-vs-E0	M2	Rem
Gap statement	Remark	rem:ktype-gaps	M2	Rem
Weyl factor for	Theorem	thm:weyl-factor	M4	—
K-invariance of the soliton integrand	Remark	—	M4	Rem
Mass formula prefactor	Corollary	—	M4	—
	Theorem	thm:gstar11	M5	—
P3 forces first-order time evolution	Lemma	lem:p3-markov	M5	—
Lorentz covariance selects Dirac, not Ya	Lemma	lem:p3-dirac	M5	—
C11 Step1: P3 selects the Dirac action	Proposition	prop:p3-dirac	M5	—
Exactness of WKB for action on	Proposition	prop:g1-dh	M4	Proved
G1 factor: closure of the WKB-to-Helgaso	Corollary	cor:g1	M4	Proved
G1 Spinorial Period from NS Boundary Con	Lemma	lem:g1-spinorial	M4	Proved
(remark)	Remark	—	M4	Rem
Compact Lie groups have no non-compact c	Lemma	lem:compact-noncompact	M2	—
does not embed in any compact group	Lemma	lem:lorentz-noncompact	M2	—
Explicit embedding of into	Lemma	lem:lorentz-embeds	M2	—
Real Form Forced by P3	Theorem	thm:realform-p3	M2	Proved
ID1 Complete	Corollary	cor:id1-complete	M2	—
Uniqueness of the QCT Physical Hilbert S	Theorem	thm:ym-hilbert-unique	M3	—
Existence	Theorem	thm:ym-hilbert-exist	M3	—
cor:hilbert-unique	Corollary	cor:hilbert-unique	M3	—
Dispersion Relation	Theorem	thm:dispersion	M3	—
Tower Bound	Theorem	thm:ff-tower	M3	—

Name / Title	Type	Label	Mod	Status
Mass Gap Independence from LSZ	Theorem	thm:lsz-independent	M3	—
QCT-Native Continuum Edge	Theorem	thm:bf-continuum	M3	—
Uniqueness of the Canonical Quantization	Theorem	thm:canon-quant-unique	M3	—
QCT Yang–Mills Mass Gap	Theorem	thm:ym-mass-gap-full	M3	Proved
Group dependence of the numerical gap va	Remark	—	M3	Rem
GKdim Path to	Theorem	—	M3	—
Algebraic Identity	Theorem	—	M3	—
Joseph Ideal Path to	Theorem	—	M3	—
(remark)	Remark	—	M3	Rem
Zero-mode count	Theorem	—	M3	—
: Hopf action	Theorem	—	M3	—
Empirical fingerprint	Remark	—	M3	Rem
Partition function $Z[0]$	Theorem	—	M3	—
	Theorem	—	M3	—
Combinatorial lower bound	Lemma	—	M3	—
Space separation	Theorem	—	M3	—
Homotopy invariance	Theorem	—	M3	—
Integrality of discrete	Theorem	—	M3	—
is more than sufficient	Remark	—	M3	Rem
Bergman correction	Theorem	—	M3	—
Proton on	Proposition	—	M3	—
Proton Prefactor	Theorem	—	M3	—
	Theorem	—	M3	—
Proton tree-level mass ratio, proved	Theorem	thm:proton-tree-proved	M3	—
Two-loop correction coefficient	Theorem	thm:proton-alpha2	M3	—
Rac and Di,	Definition	—	M3	Def
Pseudohermitian Chern invariant	Definition	—	M3	Def
TW invariant of	Theorem	thm:tw-invariant	M3	—
(remark)	Remark	—	M3	Rem
Rumin-normalised volume	Definition	—	M3	Def
Rumin volume of ,	Lemma	—	M3	—
Muon mass ratio, tree level	Theorem	thm:muon-tree	M3	—
Root-channel theorem	Theorem	thm:root-channel	M3	—
(corollary)	Corollary	—	M3	—
Muon mass formula, one-loop	Theorem	thm:muon-oneloop	M3	—
Photon-overlap equality	Theorem	thm:overlap	M3	—
The ratio is exact at one loop	Corollary	cor:onethird	M3	—
Seeley-DeWitt geometric derivation of th	Remark	—	M3	Rem
(Yang-Mills Existence and Mass Gap)	Theorem	—	M3	—

Name / Title	Type	Label	Mod	Status
(Clay)	Corollary	—	M3	—
OS2: Euclidean covariance via direct arg	Lemma	lem:OS2-direct	M3	—
(remark)	Remark	—	M3	Rem
OS3: Reflection Positivity for Bounded S	Lemma	lem:OS3	M3	—
Comparison with the compact-group case	Remark	rem:OS3comparison	M3	Rem
-model identifies with Yang–Mills	Proposition	prop:sigma-ym	M3	—
Explicit Bergman–Faddeev–Popov identification	Proposition	prop:jberg-fp	M3	Proved
$(J_{\text{Berg}} \rightarrow \det(M_{\text{FP}}))$				
Quantum stability of T5.1: no relevant	Lemma	lem:t51q	M3	Proved
(All-orders Symanzik improvement)	Remark	—	M3	Rem
Rac ground state energy equals	Proposition	prop:rac-mass	M3	—
On the Casimir eigenvalue	Remark	—	M3	Rem
Physical Hamiltonian	Remark	rem:H-vs-C2	M3	Rem
vs.Casimir operator				
Ground-band isomorphism	Proposition	prop:ground-band-iso	M3	—
(GKdim gap for)	Proposition	—	M3	—
(Mosco convergence strong resolvent conv	Proposition	—	M3	—
(Topological lower bound on the gap)	Proposition	—	M3	—
()	Remark	—	M3	Rem
T5 (complete)	Theorem	—	M3	Proved
Metric Isometry	Theorem	—	M3	—
(remark)	Remark	—	M3	Rem
QCT Hubble-Sphere Mass	Lemma	lem:mass-formula	M3	—
Schwarzschild Radius at Epoch	Corollary	—	M3	—
QCT Horizon Schwarzschild Horizon	Proposition	—	M3	—
Numerical Entropy	Lemma	—	M3	—
Surface Gravity and Temperature	Lemma	—	M3	—
Numerical Temperature and Parent Mass	Lemma	—	M3	—
QCT Boundary is a One-Way Horizon	Proposition	prop:one-way	M3	—
Thermodynamic Correspondence	Theorem	thm:thermo	M3	—
Causal Equivalence, Theorem U1-C	Theorem	thm:causal	M3	—

Name / Title	Type	Label	Mod	Status
(corollary)	Corollary	—	M3	—
Phase1 Does Not Modify the TB Equation	Lemma	lem:phase1-TB	M3	—
Schwarzschild Radius Equals Hubble Radiu	Proposition	prop:tautology	M3	—
Theorem U1 with Phase1 Cor- rection	Theorem	thm:U1-full	M3	—
Misner–Sharp Mass, Radiation Era	Lemma	—	M3	—
Misner–Sharp Mass, Matter Era (corollary)	Lemma Corollary	— —	M3 M3	— —
Universe Age with (QCT C6 Prediction)	Lemma	—	M3	—
Universe Age with (Observed) (remark)	Lemma Remark	— —	M3 M3	— Rem